

Mutual Fund Analytics Dashboard

Capstone Project — Final Report (v1.0 · June 2026)

40 Schemes · 10 Fund Houses · 12 Categories · Jan 2022 – Dec 2025

ETL Pipeline · EDA · Risk-Adjusted Metrics · Interactive Dashboard

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Dataset: 40 Schemes · 10 AMCs · 62.74 Lakh Crore AUM · 2022–2026

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SECTION 1

Executive Summary

This report presents findings of the Bluestock Mutual Fund Analytics Capstone. **All figures are derived from actual uploaded datasets** - no values have been assumed or fabricated.

40	10	12	62.74L Cr
Schemes	Fund Houses	Categories	Total AUM
1.36	2.08	31,002 Cr	26.12 Cr
Avg Sharpe	Avg Sortino	Peak SIP Dec'25	Total Folios

Key Highlights:

- Small Cap funds: highest avg 3-yr return 21.69% | Gilt: lowest at 5.69%
- SIP inflows grew 2.7×: 11,517 Cr (Jan 2022) → 31,002 Cr (Dec 2025)
- Folio count doubled: 13.26 Crore → 26.12 Crore (Jan 2022 → Dec 2025)
- Top 3 AMCs (SBI 12.5L Cr, ICICI 10.74L Cr, HDFC 9.3L Cr) = 51.8% of AUM
- Avg Sharpe 1.36 | Avg Sortino 2.08 | Worst Max Drawdown –33.5% (SBI Small Cap Direct)
- NAV correlations near zero (–0.09 to +0.05): strong diversification benefits confirmed
- Investors: 41.1% aged 26–35 | 66.5% male | T30 cities 66.3% | Lumpsum 58.49%

SECTION 2

Data Sources & Quality

10 datasets were used. Data Quality Report (Day 1) confirmed: no duplicates, all 40 AMFI codes validated, overall quality GOOD. Only 12 missing values found in yoy_growth_pct of monthly SIP file.

Dataset	Key Content	Quality
01_fund_master.csv	40 schemes, 10 AMCs, 12 categories	✓ Clean
02_nav_history_cleaned.csv	~58,400 daily NAV rows	✓ Clean
03_aum_by_fund_house.csv	Quarterly AUM 2022–2025 by AMC	✓ Clean
04_monthly_sip_inflows.csv	48 months SIP data	12 nulls in yoy_growth
05_category_inflows.csv	Monthly net inflows by category	✓ Clean
06_industry_folio_count.csv	16 monthly folio snapshots	✓ Clean
07_scheme_performance_cleaned.csv	40 rows, 20 performance cols	✓ Clean
08_investor_transactions_cleaned.csv	10,000+ transactions, 13 cols	✓ Clean
09_portfolio_holdings.csv	Sector allocation data	✓ Clean
10_benchmark_indices.csv	~1,460 rows benchmark data	✓ Clean

SECTION 3

ETL Pipeline Design

The pipeline runs through 5 stages via run_pipeline.py - a single CLI entry point.

Extract:

AMFI REST API (mfapi.in) for NAV history. Monthly AMFI reports for AUM/SIP/folio data. Live NAV for 5 key schemes. Concurrent fetching with retry logic.

Transform:

NAV Cleaning: Forward-fill, align to trading calendar, remove anomaly_flag=True rows

Return Computation: 1yr/3yr/5yr CAGR; rolling volatility windows

Risk Metrics: Sharpe, Sortino, Alpha, Beta, Max Drawdown per scheme

Fund Scorecard: Composite rank from CAGR+Sharpe+Alpha+MDD (equal weight)

Load - SQLite (bluestock_mf.db):

Table	Rows	Description
scheme_master	40	Master: amfi_code, name, AMC, category, plan
nav_history	~58,400	Daily NAV (raw + pct return)
scheme_performance	40	All 20 computed performance columns
fund_scorecard	~25	Composite scores and ranks
aum_by_fund_house	~40/quarter	Quarterly AUM snapshots

SECTION 4

EDA Findings

4.1 Fund Universe & Category Analysis

Category	N	Avg 3-Yr Return	Avg Sharpe
Small Cap	6	21.69%	0.87
Mid Cap	7	16.59%	0.87
Flexi Cap	2	15.50%	0.97
Value	1	14.76%	0.98
Large & Mid Cap	1	14.56%	0.91
ELSS	1	13.58%	0.80
Large Cap	14	12.99%	0.93
Index/ETF	2	11.94%	0.92
Short Duration	1	7.37%	1.84
Liquid	3	6.33%	6.33
Gilt	2	5.69%	1.43

4.2 AUM & SIP Trend Analysis

Total AUM reached **62.74 Lakh Crore (Dec 2025)** from ~30 Cr in 2022.

Fund House	AUM Dec'25 ('000 Cr)	Share
SBI Mutual Fund	1,250	19.9%
ICICI Prudential MF	1,074	17.1%
HDFC Mutual Fund	930	14.8%
Nippon India MF	700	11.2%
Kotak Mahindra MF	580	9.2%
Aditya Birla Sun Life MF	460	7.3%
UTI Mutual Fund	410	6.5%
Axis Mutual Fund	350	5.6%
Mirae Asset MF	290	4.6%
DSP Mutual Fund	230	3.7%

SIP inflows grew from █11,517 Cr (Jan 2022) to █31,002 Cr (Dec 2025). Active SIP accounts: 4.91 Cr → 9.35 Cr. Folio count: 13.26 Cr → 26.12 Cr. Liquid funds led category inflows at 451,275 Cr (FY25).

4.3 Investor Demographics (08_investor_transactions_cleaned.csv)

Dimension	Value	Insight
Top Age Group	26–35 (41.1%)	Largest segment; followed by 36–45 (24.9%)
Gender	Male 66.5% / Female 33.5%	Significant growth headroom for female investors
City Tier	T30: 66.3% / B30: 33.7%	B30 cities are an emerging growth market
Transaction Type	Lumpsum 58.49%, Redemption 35.34%, SIP 6.17%	Lumpsum dominant in dataset
Top State (SIP Vol)	Madhya Pradesh (~2.07 Cr)	Closely followed by Punjab (2.01 Cr)
Top Sector (Portfolio)	Banking 19.2%	IT 13.4%, Pharma 12.0%, Auto 9.5%

SECTION 5

Performance Analysis

5.1 Risk-Adjusted Metrics (07_scheme_performance_cleaned.csv)

1.36	2.08	1.25	0.87	−33.5%
Avg Sharpe	Avg Sortino	Avg Alpha	Avg Beta	Worst Max DD

Avg Sharpe of 1.36 exceeds the standard 1.0 threshold. Among equity-only funds, Sharpe ranges from −0.21 to 1.45. Avg Beta of 0.87 indicates slightly defensive positioning. Risk grade: 40% Moderate, 20% High, 15% Very High, 15% Low, 10% Moderately High.

5.2 Fund Rankings (fund_scorecard.csv + cagr_table.csv)

Ran k	Fund	Category	CAGR	Sharpe	Alpha	Max DD
1	ICICI Pru Midcap – Regular	Mid Cap	32.8%	—	—	—
2	SBI Small Cap – Regular	Small Cap	32.4%	0.94	1.23	−13.35%
3	DSP Small Cap – Regular	Small Cap	32.26%	—	—	—
4	Mirae Tax Saver – Regular	ELSS	31.92%	1.23	0.01	−16.4%
5	Mirae Large Cap – Regular	Large Cap	30.95%	1.45	−0.01	−11.27%
6	Axis Midcap – Regular	Mid Cap	28.19%	1.00	0.03	−20.96%
7	ABSL Small Cap – Regular	Small Cap	22.38%	0.90	1.84	—
8	Axis Small Cap – Regular	Small Cap	20.98%	0.84	0.51	—

5.3 Benchmark & Correlation

NAV Return Correlation Matrix (10 funds): most pairwise values between −0.09 and +0.05. Near-zero cross-fund correlations confirm genuine diversification. Sector allocation: Banking 19.2%, IT 13.4%, Pharma 12.0%, Automobile 9.5% - top 4 = 54.1% of holdings.

SECTION 6

Dashboard Architecture

Four interactive dashboards were built (PDFs included in repository):

Dashboard 1 - MF Performance Analysis:

Risk vs Return scatter (Std Dev vs 3yr Return, bubble=AUM). Avg 3-Year Return by category bar chart. Top Performing Funds table.

Dashboard 2 - SIP & Market Trends:

SIP Inflows vs Nifty 50 dual-axis line (2022–2026). Category Net Inflow Heatmap. Top 5 Categories FY25.

Dashboard 3 - MF Industry Overview:

KPIs: 13,698 schemes, 39.18M AUM, 939.72K SIP, 416.42 Cr folios. AUM by AMC bar chart. Industry AUM trend (2022–2025).

Dashboard 4 - Investor Analytics:

Transactions by state. Transaction split donut. Avg SIP by age group. Monthly transaction volume by type.

SECTION 7

Limitations

- **Sample:** 40 schemes (10 AMCs) vs 3,000+ AMFI universe - representative but not exhaustive
- **Survivorship bias:** Only active schemes; wound-up funds excluded
- **CAGR window:** cagr_table.csv uses full NAV history, not fixed 5-year window
- **Benchmark:** All equity funds benchmarked against Nifty 500; thematic funds may differ
- **Batch pipeline:** No real-time streaming; data current as of last run
- **Transaction data:** Investor dataset is simulated for this internship

SECTION 8

Recommendations

For Investors:

- Filter by Sharpe > 1.0 - 40% of our 40-scheme universe qualifies
- Small Cap and Mid Cap offer best 3-yr returns (21.69% and 16.59%) with moderate Sharpe
- Liquid funds suitable for short-term - 6.33% avg return, near-zero volatility
- Near-zero cross-fund correlations confirm multi-category diversification benefits

Platform v2.0 Roadmap:

- Expand to full 3,000+ scheme universe via complete AMFI API crawl
- Implement daily refresh via GitHub Actions cron job
- Add SIP XIRR calculator and goal-based planning module
- Build ML scoring model (XGBoost on Sharpe, Alpha, MDD, CAGR features)
- Deploy dashboard on Streamlit Cloud or GitHub Pages

SECTION 9

Conclusion

The Bluestock MF Analytics Capstone delivered all Day 7 deliverables using real AMFI and industry data. Key finding: the Indian MF industry showed remarkable growth 2022–2025, with SIP inflows tripling, folios doubling, and AUM reaching 62.74L Cr. Small and Mid Cap equity funds led performance. Near-zero cross-fund correlations confirm genuine diversification value within this universe.

APPENDIX

Appendix A - Dataset Schema

07_scheme_performance_cleaned.csv

Column	Type	Description
amfi_code	INT	AMFI scheme identifier
scheme_name	TEXT	Full scheme name
fund_house	TEXT	AMC name
category	TEXT	SEBI category
plan	TEXT	Direct/Regular
return_1yr_pct	FLOAT	1-year return %
return_3yr_pct	FLOAT	3-year return %
return_5yr_pct	FLOAT	5-year return %
sharpe_ratio	FLOAT	Sharpe ratio
sortino_ratio	FLOAT	Sortino ratio
alpha	FLOAT	Jensen's Alpha
beta	FLOAT	Market beta
max_drawdown_pct	FLOAT	Max drawdown %
aum_crore	FLOAT	AUM Crore
risk_grade	TEXT	Moderate/High/Very High/Low/Moderately High
anomaly_flag	BOOL	True if outlier detected

APPENDIX

Appendix B-Metric Formulae

CAGR: $CAGR = (NAV_end / NAV_start)^{(1/n)} - 1$

n = years; used for 1yr, 3yr, 5yr

Sharpe Ratio: $Sharpe = (R_p - R_f) / \sigma_p$

R_f = 6.5%; σ_p = annualised std dev of daily returns

Sortino Ratio: $Sortino = (R_p - R_f) / \sigma_d$

σ_d = downside deviation (negative returns only)

Jensen's Alpha: $\alpha = R_p - [R_f + \beta \times (R_m - R_f)]$

R_m = Nifty 500 TRI; β from OLS regression

Beta: $\beta = Cov(R_p, R_m) / Var(R_m)$

3-year rolling window of daily returns

Max Drawdown: $MaxDD = (Trough - Peak) / Peak$

Peak = running max NAV; Trough = min after peak

Fund Scorecard: $Score = \text{mean}(CAGR_rank, Sharpe_rank, Alpha_rank, MDD_rank)$

Equal-weight composite; higher = better